Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Object

Meaning in my words In a force diagram, what do we mean by the object?	**How can you tell?** You circle only the cart. Which force belongs: the hand on the cart or the cart on the hand?
Example or sketch What would you put at the centre of a free-body diagram for a hanging lamp?	**Non-example and why** A student switches to forces on the ceiling halfway through. How should they organize those?

External force

Meaning in my words How would you explain external force to a friend?	**How can you tell?** Draw a boundary around a lamp alone. Why is the cable's pull external?
Example or sketch If your system contains two carts and a hand pushes one from outside, what force would you include?	**Non-example and why** If lamp and cable are both inside the boundary, should their mutual pull count as an external force?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Force-arrow scale

Meaning in my words	How can you tell?
How would you explain force-arrow scale to a friend?	Your scale is 1 cm for 2 N. How long should a 6 N arrow be?
Example or sketch	**Non-example and why**
Draw 2 N left and 6 N right at that scale.	Two arrows of equal length are labeled 2 N and 8 N on one diagram. What would you change?

Net force

Meaning in my words	How can you tell?
How would you explain net force to a friend?	Two people push one cart in opposite directions. Why do the arrow directions matter?
Example or sketch	**Non-example and why**
One person pushes 6 N right and another 4 N left. What single arrow replaces both?	A student adds those pushes to get 10 N right. Which push did they count the wrong way?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch the lamp, its boundary, and its two force arrows. Make their lengths predict upward acceleration.

Chosen object: _____

My marks mean

On assignment: External force

SECTION E / BACK / FINAL MEETING ONLY Name _____

The mission simulator needs one overlay called “forces on XJ.” Its proposed layers include Earth on XJ, XJ on Earth and XJ’s velocity.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Systems engineer**

Choose the boundary of the simulated object. Draw only the force layers you would send to the acceleration calculation.

☐ **Animation director**

Make a two-layer storyboard: physical forces on XJ, and motion. Tell the animator which Earth arrow belongs in another shot.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor’s check: Distinguish the net-force direction from a possible velocity direction.